

42
FORM TP 2009183



TEST CODE 02220020

MAY/JUNE 2009

CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE

UNIT 2: AGRICULTURE, ENERGY AND ENVIRONMENTAL POLLUTION

PAPER 02

2 hours 30 minutes

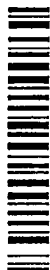
22 MAY 2009 (p.m.)

COUVA EAST SECONDARY
LIBRARY
636-2585

C.E.S.E.C.

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. **DO NOT** open this examination paper until instructed to do so.
2. This paper consists of **NINE** questions, **THREE** from each Module.
3. Candidates **MUST** answer **SIX** questions, **TWO** from **EACH** Module.
4. Write your answers in the answer booklet provided.
5. The use of a silent, non-programmable, scientific calculator is permitted.



MODULE 1

Answer ANY TWO questions.

1. (a) State TWO characteristics of:

(i) Intensive aquaculture

[2 marks]

(ii) Extensive aquaculture

[2 marks]

(b) Study Figure 1 which shows the annual production of fish protein from traditional marine fishing and from the non-traditional practice of aquaculture.

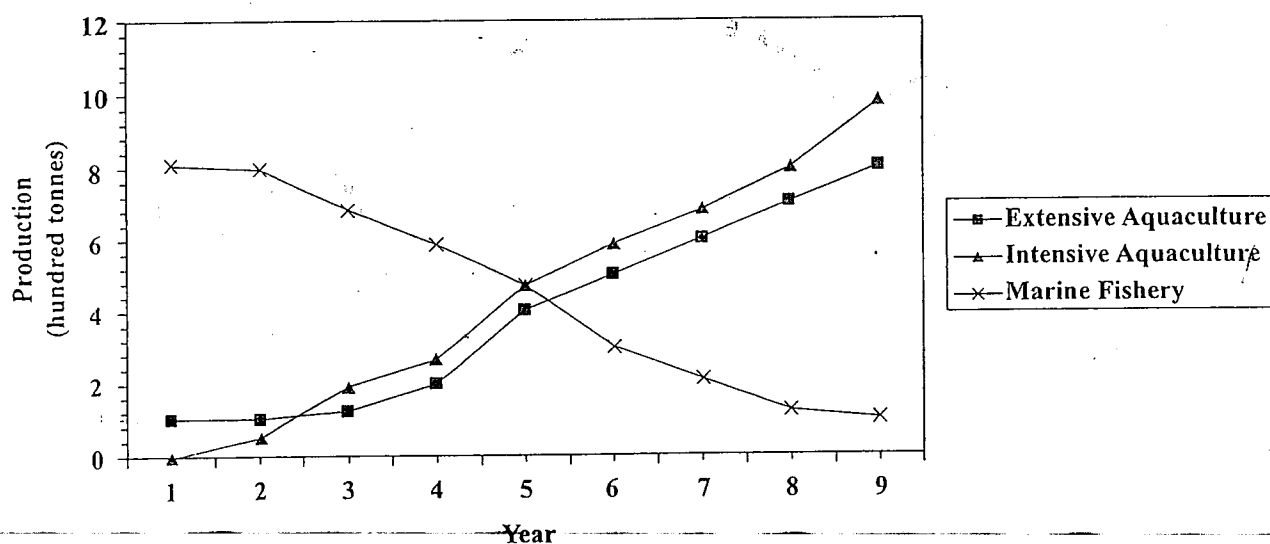


Figure 1. Production of fish protein from traditional and non-traditional fisheries

(i) Describe the trend in fish protein production illustrated in Figure 1. Include specific values in your response. [4 marks]

(ii) Which of the two aquaculture methods shown in Figure 1 would you expect to have a GREATER impact on the environment? Suggest TWO reasons to justify your response. [6 marks]

(c) Discuss TWO environmental concerns associated with the practice of aquaculture. [6 marks]

Total 20 marks

GO ON TO THE NEXT PAGE

2. (a) Table 1 shows the results of using two alternative pest control strategies during the cultivation of rice in two similar 10-hectare fields.

TABLE 1: PEST CONTROL RESULTS

Parameter	Use of Chemical Pest Control Method	Use of Integrated Pest Control Method
Number of times pesticide used	9	1
Cost (per hectare per application)	\$3000	\$8000
Rice yield (bags per hectare)	50	60
Diversity of species in rice field after application of method	1.5	4.5
Ease of application	Easy	Difficult

- (i) Which of the two methods of pest control shown in Table 1 would you recommend to rice farmers? Justify your response. [6 marks]
- (ii) Describe **another** environmentally sustainable pest control method that is available to rice farmers. [4 marks]
- (b) Mr. Farmer lamented to his wife that whenever it rains the water runs downhill very quickly and the more quickly it runs, the more soil it carries off his hillside farm.
- (i) Recommend TWO environmentally sustainable practices in agricultural systems that Mr. Farmer can use to reduce water runoff and erosion on his hillside farm. [2 marks]
- (ii) Justify EACH of the environmentally sustainable practices recommended in (b) (i) above. [8 marks]

Total 20 marks

GO ON TO THE NEXT PAGE

3. (a) Study Figure 2 which shows an agricultural system practised by Farmer Smith and answer the questions below.

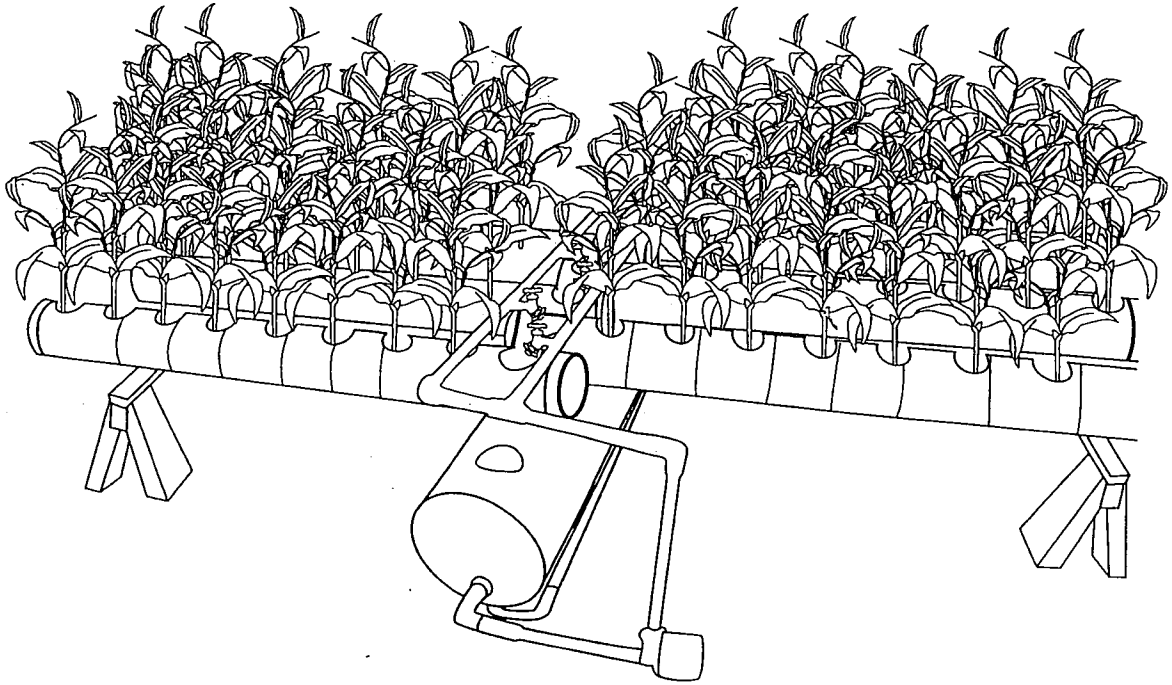


Figure 2. An agricultural system practised by Farmer Smith

- (i) Identify the agricultural practice shown in Figure 2. [1 mark]
- (ii) Outline FOUR features of this agricultural practice. [4 marks]

GO ON TO THE NEXT PAGE

- (b) An insert is provided for this question.

Table 2 presents a record of the yield of tomatoes harvested by Farmer Smith for the period 2000 – 2008. Plot an appropriate graph on the insert provided, to illustrate the results shown in Table 2.

TABLE 2: RESULTS FOR ANNUAL YIELD

Year	Annual Yield (thousand kg)
2000	6
2001	6
2002	7
2003	9
2004	12
2005	15
2006	18
2007	22
2008	26

[5 marks]

- (c) Suggest FOUR reasons for the trend observed in the graph plotted in Part (b) above.
[4 marks]

- (d) Farmer Smith believes his method of cultivation is sustainable. Suggest THREE reasons why this may be so.
[6 marks]

Total 20 marks

COUVA EAST SECONDARY
LIBRARY
636-2585

C.E.S.E.C.

GO ON TO THE NEXT PAGE

MODULE 2

Answer ANY TWO questions.

4. (a) Study Figure 3 which shows the distribution of fossil fuel use, in **million metric tonnes of oil equivalents**, for two Caribbean countries, A and B, and answer the questions below.

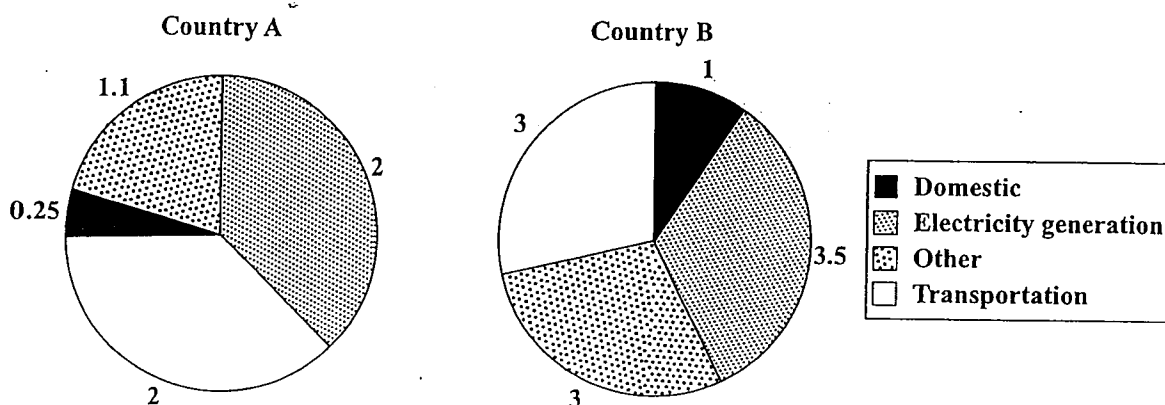


Figure 3. Distribution of fossil fuel use in two countries

- (i) Determine the quantity of fossil fuel used for domestic and transportation purposes in EACH country. [2 marks]
- (ii) Determine which country uses the larger percentage of its fossil fuel for electricity generation. [3 marks]
- (iii) State THREE uses that could be included in the "Other" category shown in Figure 3. [3 marks]
- (b) The government of a Caribbean country is planning to set up a new industrial estate in a remote area. Discuss FOUR factors that would affect the supply of energy to the new industrial estate. [12 marks]

Total 20 marks

GO ON TO THE NEXT PAGE

5. (a) Study Figure 4 which shows the cost of electricity from different sources of energy and answer the questions below.

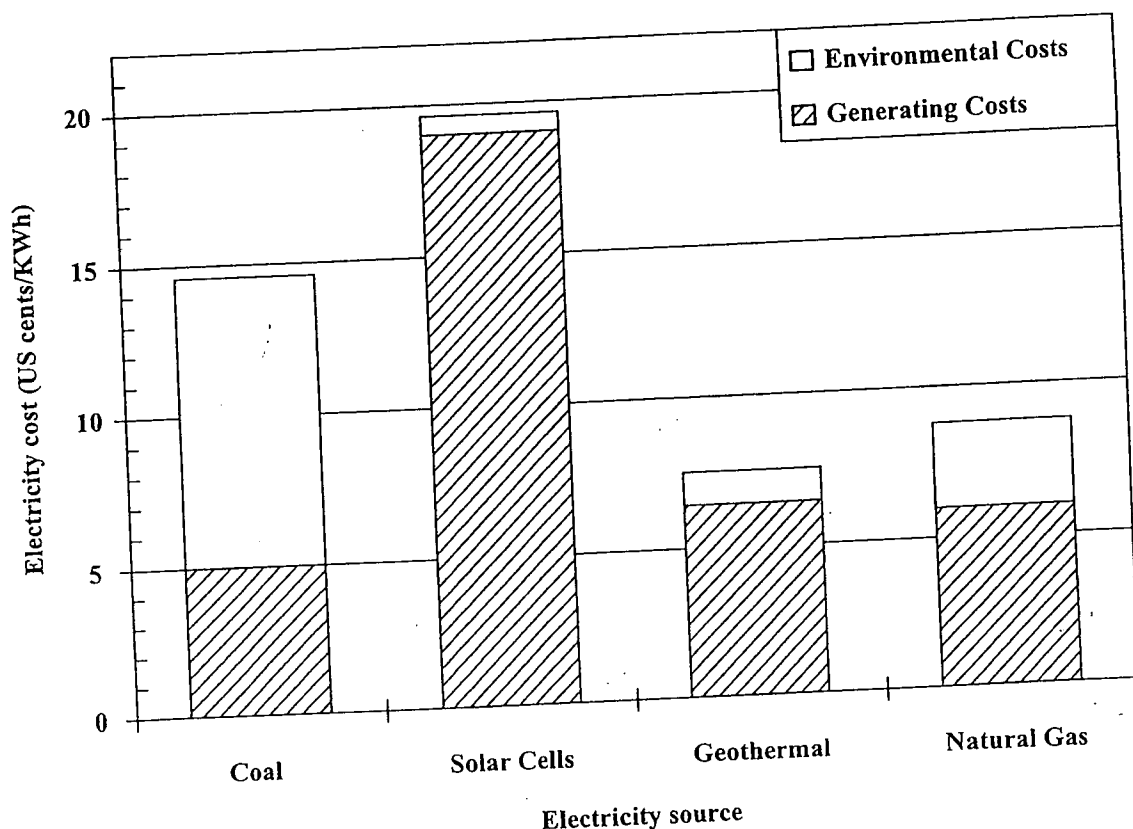


Figure 4. Cost of electricity from different sources of energy

- (i) Determine the environmental cost of using natural gas as a source for electricity production. [2 marks]
- (ii) A country is considering switching from using coal to either geothermal energy or solar cells as a source for electricity production. Based SOLELY on the data in Figure 4, which form of renewable energy would you recommend? Justify your choice with TWO reasons, using specific values. [6 marks]
- (b) List THREE other factors, EXCEPT cost, that must be considered for successfully implementing the switch from coal to the form of renewable energy that you recommended in Part (a) (ii). [3 marks]
- (c) For a NAMED Caribbean country, indicate whether the renewable energy recommendation you made in Part (a) (ii) would be a suitable replacement for coal. Provide THREE reasons to justify your recommendation. [9 marks]

Total 20 marks

GO ON TO THE NEXT PAGE

6. (a) Study Figure 5 which illustrates the changing requirements for nuclear power plants in the United States of America between 1964 and 1995.

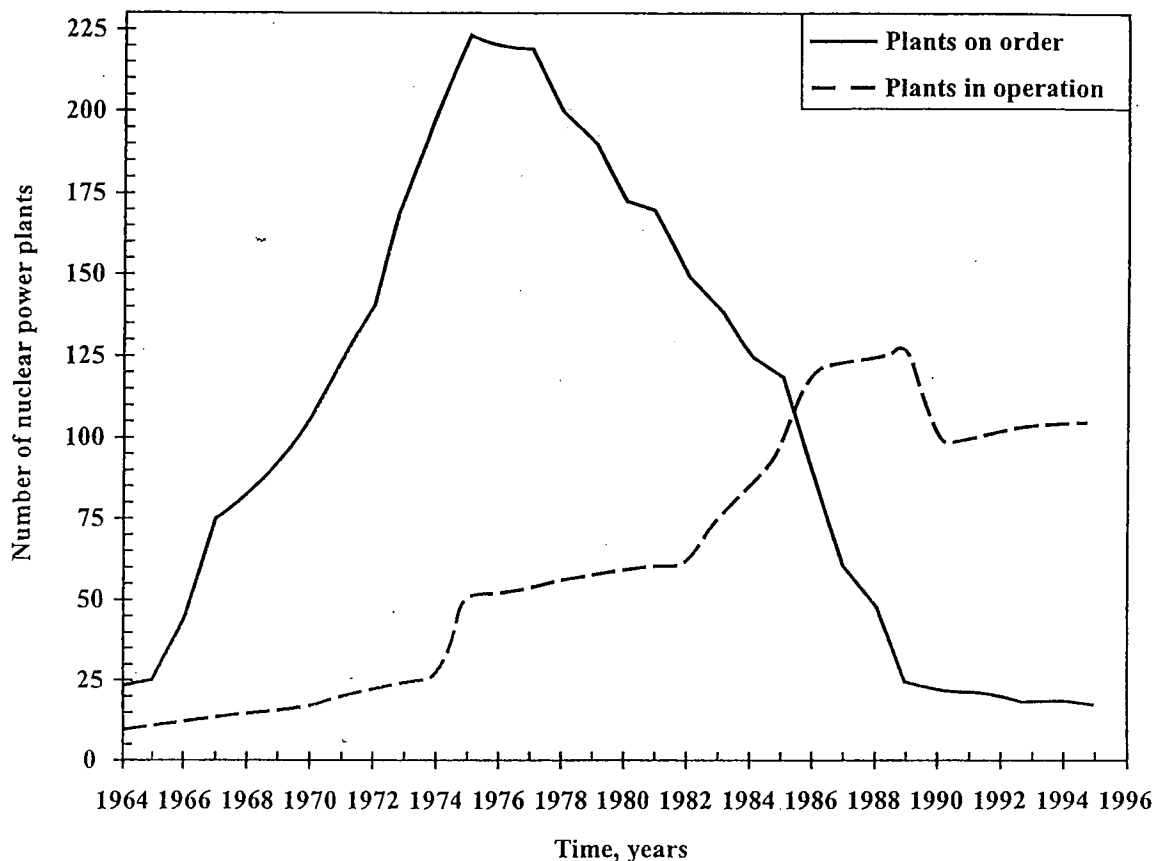


Figure 5. Illustration of the changing requirements for nuclear power plants in the United States between 1964 and 1995

- (i) In which year was the number of plants on order equal to the number of plants in operation? [1 mark]
 - (ii) In which year was the HIGHEST number of nuclear power plants on order? [1 mark]
 - (iii) How many nuclear power plants were on order in 1995? [1 mark]
 - (iv) What was the average rate of decline in the number of nuclear power plants on order over the period shown in Figure 5? [2 marks]
- (b) Discuss the impact of the use of nuclear power on the environment. Your answer should include TWO positive and THREE negative impacts. [15 marks]

Total 20 marks

GO ON TO THE NEXT PAGE

MODULE 3

Answer ANY TWO questions.

7. (a) Study Table 3 which shows the amounts and distribution of solid waste disposed at a landfill daily in Trinidad and answer the questions below.

TABLE 3: SOLID WASTE DISPOSED DAILY AT A LANDFILL IN TRINIDAD

Waste Type	Mass (tonnes/day)
Metal	0.20
Other/Glass	0.28
Paper products	0.60
Plastic	0.48
Textile	0.16
Wood	0.8
Yard and food waste	2.20

- (i) Rank the different waste types in order of INCREASING daily mass in the landfill. [1 mark]
- (ii) Determine the **percentage** of the waste type with the SMALLEST daily mass in the landfill. [1 mark]
- (iii) How long will it take for 28 tonnes of waste of the same composition as in Table 3 to be disposed of in the landfill? [2 marks]
- (iv) Suggest ONE reason why the amount of metal disposed of in the landfill is low. [2 marks]
- (b) (i) Define the term 'leachate'. [2 marks]
- (ii) Identify FOUR pollutants, EXCEPT leachate, that can be released by a landfill. [4 marks]

GOUVA EAST SECONDARY
LIBRARY
636-2585

C.E.S.E.C.

GO ON TO THE NEXT PAGE

- (c) Figure 6 illustrates an area around a landfill. Water extracted from both wells was tested and only one of the drinking water wells in the area was found to be contaminated by a landfill pollutant.

Study Figure 6 and answer the questions below.

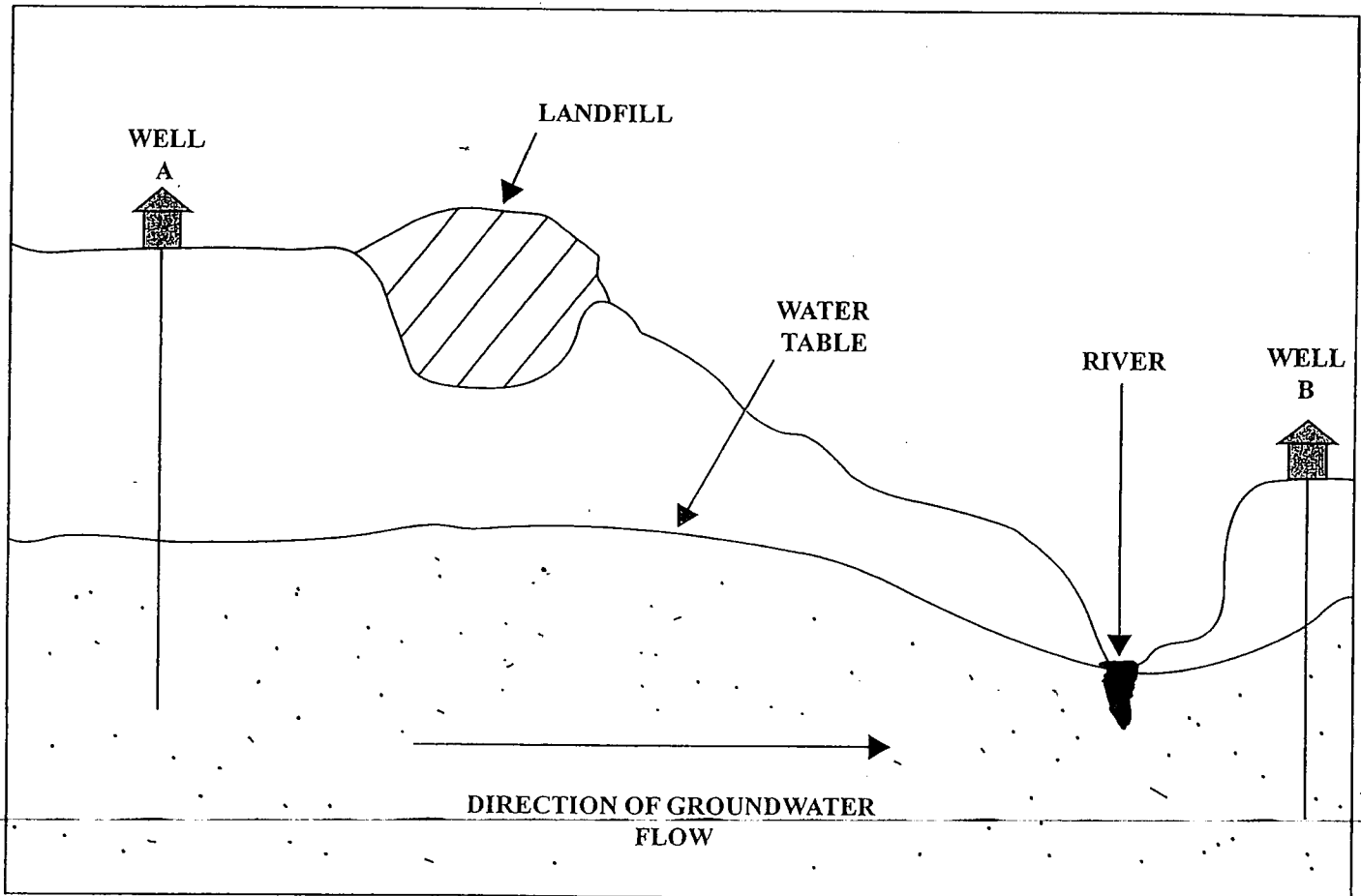


Figure 6. Diagram of an area containing a landfill

- (i) Which well is MOST likely to be contaminated? [2 marks]
- (ii) Explain how a pollutant from the landfill could contaminate the drinking water well. [4 marks]
- (iii) Why is the other well NOT likely to be contaminated? [2 marks]

Total 20 marks

8. (a) Define the term 'noise pollution'. [2 marks]
- (b) Explain why noise is considered as a pollutant. Provide THREE reasons in your answer. [6 marks]
- (c) Study Figure 7 which illustrates the results of a noise monitoring exercise at a carnival fete and answer the questions below.

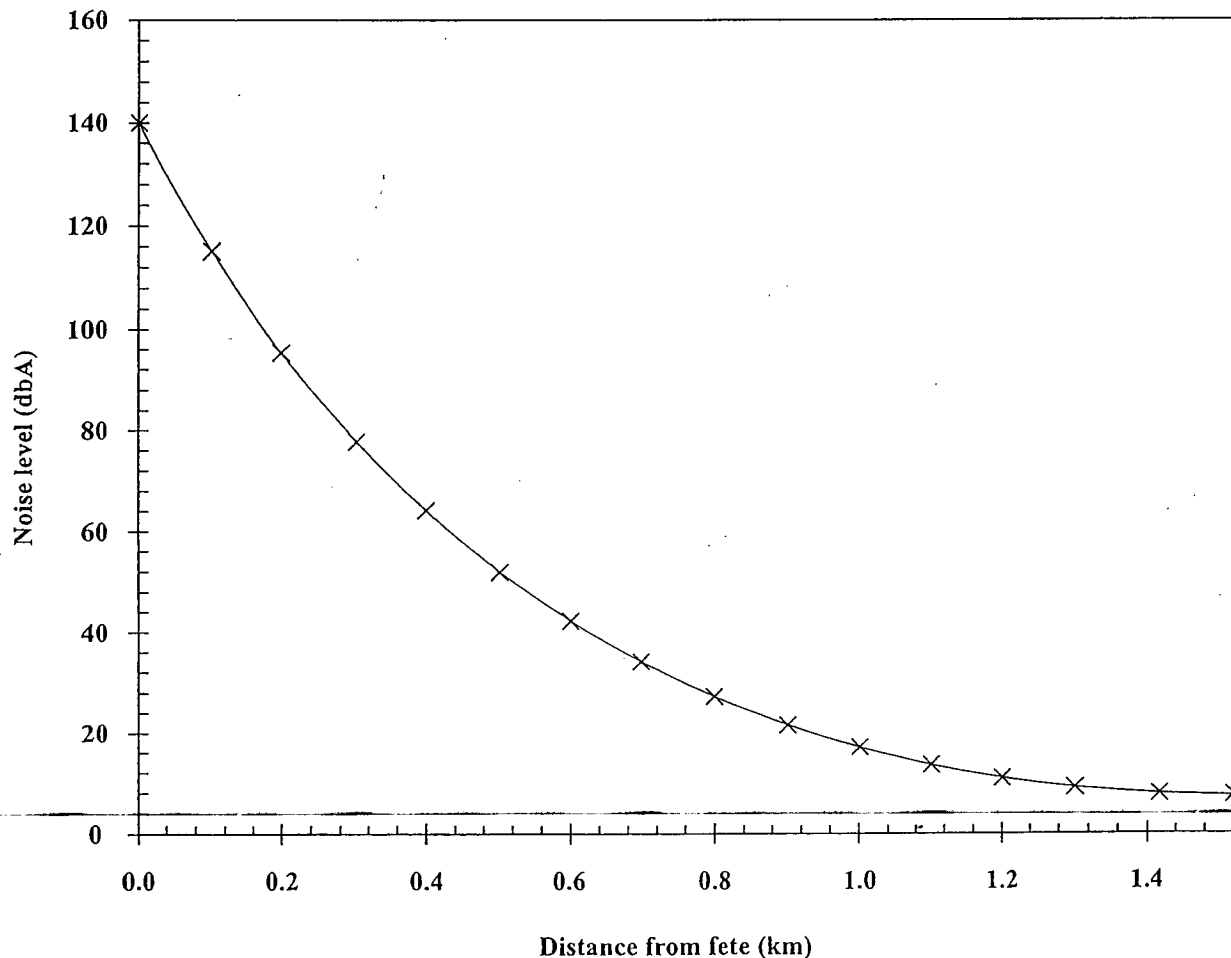


Figure 7. Results of a noise monitoring exercise at a carnival fete

- (i) At what distance is it **likely** that a homeowner would NOT be disturbed by the noise from the fete? [2 marks]
- (ii) A party-goer had to park his car 0.5 km away from the fete. How far would he have to walk before he began to experience a harmful noise level of 84 dbA? [2 marks]
- (d) Explain how any TWO measures reduce noise pollution. In your response, state ONE example of how EACH measure is used. [8 marks]

Total 20 marks

GO ON TO THE NEXT PAGE

9. (a) **Figure 8 is provided as an insert.**

Study Figure 8 which is a diagram showing ozone concentration in the earth's atmosphere and answer the questions below.

- (i) Label any TWO layers of the atmosphere. [2 marks]
 - (ii) On Figure 8, indicate the position of the ozone layer. [1 mark]
 - (iii) At what altitude is the ozone concentration LOWEST? [1 mark]
 - (iv) Describe the relationship between 'ozone concentration' and 'altitude'. [3 marks]
- (b) Discuss the following statement:
- "It is vital to the good health of our environment that the ozone layer is preserved."
- Include FOUR points in your discussion. [8 marks]
- (c) Explain how a NAMED international treaty functions to protect the ozone layer and reduce ozone depletion. [5 marks]

Total 20 marks

END OF TEST

TEST CODE **02220020**

FORM TP 2009183

MAY/JUNE 2009

**CARIBBEAN EXAMINATIONS COUNCIL
HEADQUARTERS**

ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE

UNIT 2 – PAPER 02

Insert for Question 9 (a)

Centre Number _____

Candidate's Number _____

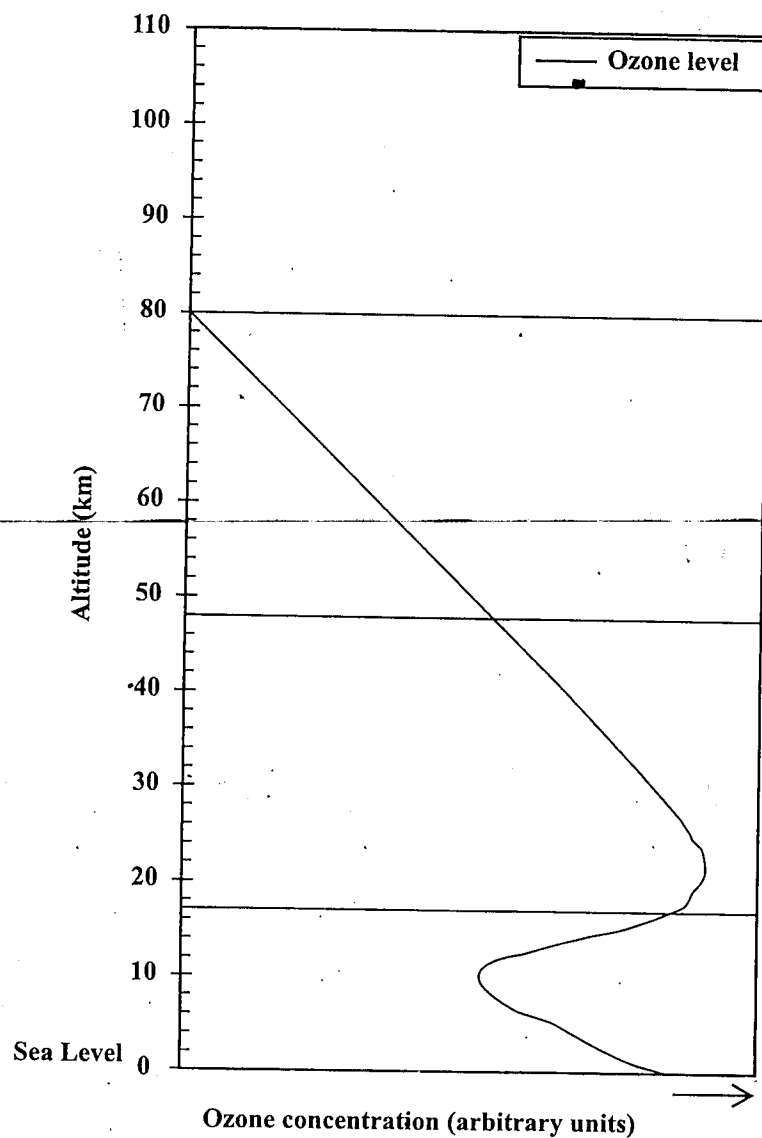


Figure 8. Diagram of the earth's atmosphere

TO BE ATTACHED TO YOUR ANSWER BOOKLET